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A constitutive modeling interpretation of the relationship among carotid artery stiffness, blood pressure, and age in hypertensive subjects

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Spronck B, Heusinkveld MH, Donders WP, de Lepper AG, Op't Roodt J, Kroon AA, Delhaas T, Reesink KD. A constitutive modeling interpretation of the relationship among carotid artery stiffness, blood pressure, and age in hypertensive subjects. *Am J Physiol Heart Circ Physiol* 308: H568–H582, 2015. First published December 24, 2014; doi:10.1152/ajpheart.00290.2014.—Aging has a profound influence on arterial wall structure and function. We have previously reported the relationship among pulse wave velocity, age, and blood pressure in hypertensive subjects. In the present study, we aimed for a quantitative interpretation of the observed changes in wall behavior with age using a constitutive modeling approach. We implemented a model of arterial wall biomechanics and fitted this to the group-averaged pressure-area (*P-A*) relationship of the "young" subgroup of our study population. Using this model as our take-off point, we assessed which parameters had to be changed to let the model describe the "old" subgroup's *P-A* relationship. We allowed elastin stiffness and collagen recruitment parameters to vary and adjusted residual stress parameters according to published age-related changes. We required wall stress to be homogeneously distributed over the arterial wall and assumed wall stress normalization with age by keeping average "old" wall stress at the "young" level. Additionally, we required axial force to remain constant over the cardiac cycle. Our simulations showed an age-related shift in pressure-load bearing from elastin to collagen, caused by a decrease in elastin stiffness and a considerable increase in collagen recruitment. Correspondingly, simulated diameter and wall thickness increased by about 20 and 17%, respectively. The latter compared well with a measured thickness increase of 21%. We conclude that the physiologically realistic changes in constitutive properties we found under physiological constraints with respect to wall stress could well explain the influence of aging in the stiffness-pressure-age pattern observed.

aging; pulse wave velocity; elastin; collagen; constitutive modeling

IT IS WELL ESTABLISHED THAT aging has a profound influence on arterial wall structure and function. Previously, we noninvasively established that in clinical hypertension patients the commonly observed increase in pulse wave velocity (PWV) with blood pressure (BP) could be directly predicted from the nonlinearity of the pressure-area (*P-A*) relationship and does not require a change of the *P-A* relationship itself (46; methods reproduced in APPENDIX). With age, however, we did find a shift of the *P-A* relationship to larger areas and an increase in steepness of the relationship, together corresponding to an increase in PWVs (46; main findings reproduced in Table 1). The BP-age-stiffness pattern is merely descriptive and does not readily reveal any mechanistic insight. The BP dependency of arterial stiffness is known to be directly linked to arterial wall matrix constitution and actual distribution of mechanical load, while age-related changes also entail adaptive responses to load. In the present study, we used a constitutive modeling approach for a quantitative interpretation of the previously observed age- and pressure-related differences in arterial stiffness. Such an approach could potentially be applicable in a broad range of clinical studies focusing on arterial wall remodeling.

Constitutive modeling of the arterial wall has been used for a variety of applications (24) such as studies of aneurysm rupture or of arterial remodeling. However, the number of studies that apply constitutive modeling to in vivo measured data is limited.

In 2003, Schulze-Bauer and Holzapfel (43) used a Fung-type combination of strain energy functions to describe the mechanical behavior of the thoracic aorta of one normotensive and one hypertensive subject. Their model, however, does not explicitly model the three major constituents of the wall, i.e., collagen, elastin, and vascular smooth muscle (VSM).

Stålhand et al. (48–50) followed a similar approach in 2004–2006, using invasively measured pressure data from a healthy volunteer. In 2009, Stålhand (47) expanded his model by including explicit isotropic and anisotropic parts, separating the anisotropic contribution of collagen from the other vascular constituents.

In 2011, Åstrand et al. (1) published a study in which they used the approach described by Stålhand in 2009 (47) to assess constitutive parameters in 30 healthy volunteers divided into 3 age groups. One of their remarkable findings was an increase of the elastin structure stiffness with age. This finding could be

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Table 1. *Previously obtained pulse wave velocities for predefined pressure ranges in relation to age*

	Pulse Wave Velocity, m/s	
	Mean age: 41 yr ("young")	Mean age: 64 yr ("old")
120 SBP/80 DBP, mmHg	7.2	10.9
160 SBP/90 DBP, mmHg	8.1	12.2

Stiffness, as indicated by pulse wave velocity (PWV), is shown for normotensive (120/80 mmHg) and hypertensive (160/90 mmHg) pressure ranges for the 2 age groups studied. Carotid artery PWV values were derived from the exponentially modeled pressure-area curves via Bramwell-Hill for the "young" and "old" groups. SBP, systolic blood pressure; DBP, diastolic blood pressure.

explained ultrastructurally by the mechanism of elastocalcinosis. Although it is established that elastocalcinosis is associated with an increased overall artery stiffness (2), it can be questioned whether this increase is due to either an increased stiffness of the load-bearing elastin or a shift in load bearing to the stiffer collagen due to elastin fiber rupture. The current paradigm of arterial aging, in which the elastin structure is thought to degrade or rupture with age (56), suggests a decrease in stiffness of the elastin structure.

Masson et al. (31) proposed a three-constituent model for clinical application in 2008. The study by Masson et al. differs from the aforementioned studies (1, 43, 47–50) in two ways. Firstly, Masson et al. (31) measured BP noninvasively using tonometry, whereas the earlier studies used invasive catheter measurements. This makes the entire data-acquisition noninvasive and increases the potential to include larger numbers of subjects. Secondly, Masson et al. assessed wall thickness patient-specifically by measuring intima-media thickness (IMT). The other studies used a population-based equation to estimate wall thickness, ignoring potential patient- and group-specific differences.

In 2011, Masson et al. (30) used their model to evaluate 16 normotensive and 25 hypertensive subjects. Although their results were promising in showing the potential of noninvasive assessment of age- and BP-induced changes in wall constituents, their study had the limitation that 14 parameters were estimated based on only the *P-A* curve. Therefore, the uniqueness of the obtained parameter values obtained is questionable.

In the present study, we modified a mathematical model developed by Zulliger et al. (64) and used this to quantitatively assess the constitutive changes with aging in the carotid artery that we became aware of due to the stiffness-BP-age pattern previously found (Table 1; Ref. 46). The model considers the arterial wall to be composed of an extracellular matrix consisting of elastin and collagen embedded with VSM, with the three constituents homogeneously distributed in the wall. VSM is assumed to be oriented/acting in circumferential direction (i.e., anisotropically), while elastin is assumed to behave isotropically. Collagen is assumed to be anisotropic and oriented in two symmetric helices, as proposed by, e.g., Holzapfel et al. (23). The model also includes parameters to incorporate changes in residual stress as they occur with aging of the arterial wall (42, 65).

We explicitly chose to do the following:

1) Focus on modeling the transition from "young" to "old" arterial wall mechanics (i.e., "aging"), fitting only six selected parameters, to avoid overfitting and related interpretation problems;

2) Investigate changes in stiffness and recruitment of the different wall constituents and resulting changes in load bearing;

3) Assess changes in wall geometry related to wall stress;

4) Use clinically obtained measurements in hypertensive patients. We measured these patients under untreated (baseline) and treated (repeat) conditions. The data thus obtained at two distinct pressure ranges allow evaluation of the consistency of the model outcomes; and

5) Use noninvasive data, including echographic estimation of wall thickness, i.e., IMT.

Our approach addresses the limitations of previous work, in which either a very large number of parameters was fitted (e.g., Ref. 30), the wall constituents were not explicitly modeled (e.g., Ref. 1, omitting VSM), or an estimated instead of a measured reference for wall thickness was used (30, 31).

We evaluate our quantitative modeling approach and discuss the potential value of constitutive model-based interpretation of clinical patient data. Explicitly, we are not proposing a new model per se but instead are focusing on interpreting clinical measurements using an established constitutive model with only minimal modifications. Such a model-based interpretation may offer insight into the individual patient's arterial biomechanical state, as needed in, e.g., stiffness-treatment studies targeting specific wall constituents.

Glossary

Patient Age Groups and Respective Models

"Young" group/model	(Model fit of) the patient group of age <50 yr
"Old" group/model	(Model fit of) the patient group of age >50 yr

Measurements

Baseline measurements	Measurements used to initially develop and fit the obtained models (at hypertensive conditions)
Repeat measurements	Measurements used to assess the reproducibility of the obtained modeling results (after hypertensive treatment)

Fitting Constraints

Wall stress homogenization	Forcing circumferential wall stress to be constant across the wall from inside to outside at mean arterial pressure
Wall stress normalization	Forcing circumferential wall stress to remain constant with age
Pressure-independence of axial force	Forcing axial force to remain constant over the cardiac cycle

MATERIALS AND METHODS

P-A Curve Data Acquisition and Processing

Data acquisition and patient data used for the present study are extensively described in the APPENDIX. The study protocol conformed

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to the Declaration of Helsinki (updated Seoul 2008) and was approved by the Maastricht University Medical Ethics Committee. Study subjects were recruited from patients attending an outpatient hypertension clinic. Measurements were performed at inclusion (baseline) and repeated after 3 mo (repeat) for each individual. *P-A* relationships were obtained at each visit (46; method summarized in APPENDIX). IMT was determined as described previously (22, 54, 60). To critically evaluate the consistency of our approach, we used baseline and repeat data from $n = 13$ patients whose BP was clearly reduced at repeat. We divided the group into a "young" group (age <50 yr; $n = 6$) and an "old" group (age >50 yr; $n = 7$). Baseline characteristics of both groups are shown in Table 2. The present analysis focuses on the differences in carotid artery wall properties between the age groups.

Constitutive Modeling

To interpret the differences in measured biomechanical behavior between age groups, we implemented a constitutive model of the arterial wall (64). We approached the group-averaged curve of the "young" patients as the reference state and, accordingly, the "old" group's data as the result of age-related changes in arterial wall matrix properties and adaptation to wall stress. We specifically quantified to which extent elastin stiffness, collagen recruitment, and VSM characteristics needed to be modified to explain the age-related differences in terms of *P-A* relation. We took into account long-term homeostasis of wall stress in the wall (1, 32, 61), independence of axial force on pressure (3, 8, 18, 26, 58), as well as changes in residual strains with age (42, 65).

Model definition. A detailed description of the model used can be found in the APPENDIX. Our model, which largely follows the model published by Zulliger et al. (64), distinguishes three wall components: elastin, collagen, and VSM. The mechanical behavior of each of these components is described by a strain energy function (SEF). SEFs provide a relationship between stress and strain and as such provide a convenient way of formulating constitutive behavior of hyperelastic materials (26).

Table 2. *Study group characteristics*

	Age Group	
	"Young" Age <50 yr	"Old" Age >50 yr
N	6	7
Sex	3f/3m	2f/5m
Age, yr	41 ± 6	64 ± 9*
Height, cm	174 ± 6	174 ± 8
Weight, kg	82 ± 18	85 ± 9
BMI, kg/m^2	27 ± 4	28 ± 2
Meds, DDD	1.0 ± 1.4	0.6 ± 1.1
SBP (P_s), mmHg	149 ± 17	174 ± 34
DBP (P_d), mmHg	95 ± 12	92 ± 4
PP (P_p), mmHg	54 ± 14	82 ± 33
A_d, mm^2	46 ± 9	59 ± 17
A_n, mm^2	50 ± 11	61 ± 17
A_s, mm^2	51 ± 11	62 ± 17
Carotid CC (C), 10^{-2} mm^2/mmHg	8.8 ± 3.2	4.8 ± 1.1*
IMT, mm	0.65 ± 0.08	0.79 ± 0.23
Carotid PWV (v_{car}), m/s	8.4 ± 1.2	12.7 ± 2.9*
Aortic PWV (v_{ao}), m/s	8.8 ± 1.0	11.1 ± 1.8*

Values are means ± SD. Meds denotes antihypertensive medication in daily defined dose (DDD). BMI, body mass index; PP, pulse pressure; A_d, A_n, and A_s, diastolic, dicrotic notch, and systolic carotid artery cross-sectional area, respectively (based on adventitia-adventitia distance); CC, compliance coefficient; IMT, intima-media thickness; v_{car}, local carotid PWV based on the Bramwell-Hill equation; v_{ao}, aortic PWV based on carotid-to-femoral transit time. Sex differences (f, female; m, male) were not statistically significant ($P = 0.59$, Fisher's exact test). *$P < 0.05$, for difference between age groups (Wilcoxon rank-sum test).

Elastin, assumed to be mechanically isotropic, is described by a convex SEF (*Eq. A4*) that is governed by parameter c_{elast}, which linearly scales the elastin stress-strain relationship.

Collagen is assumed to be oriented in two symmetric helices at an angle of $\pm\beta_0$ with respect to the circumferential orientation (see Fig. A1), similar to, e.g., Holzapfel et al. (23). The SEF of a single collagen fiber (*Eq. A8*) increases quadratically with strain. The fiber SEF is convolved with a log-logistic engagement stretch distribution (*Eq. A5*) to obtain the SEF of the collagen ensemble. The latter function describes the probability that, given a certain stretch, a collagen fiber is recruited. This distribution is governed by two parameters: *1*) b, a scaling parameter that, for increasing values, shifts the log-logistic probability density function to the right, causing collagen to engage at higher strains. At the same time, probability density function height decreases and variance increases. *2*) k, a shape parameter which, for increasing values, results in a higher maximum and a narrower distribution, thus causing collagen to engage more abruptly.

VSM is assumed to be oriented circumferentially (19, 40). The VSM SEF (*Eq. A10*) results in a linear stress-strain relationship when VSM is maximally contracted. Parameter c_{VSM} linearly scales this relationship, analogous to c_{elast} in the elastin SEF. Myogenic behavior is implemented by means of a sigmoid-shaped scaling function (S_1, *Eq. A12*). The location and dispersion of the curve are determined by parameters μ and σ, respectively.

Four additional model parameters, i.e., cross-sectional wall area (A_w), the zero-stress inner radius (R_i), the longitudinal stretch ratio (λ_z), and the opening angle (α) were used to model the "young" group data (13). Notably, we defined our opening angle from the center of the stress-free (cut open) configuration (see Fig. A1).

Age group data averaging for fitting. For each individual we fitted a single exponential (SE) function (*Eq. A3*) to the measured *P-A* data, as further detailed in the APPENDIX. To obtain group averaged *P-A* relations for the "young" group ($group$ = "young"; $n = 6$) as well as for the "old" group ($group$ = "old"; $n = 7$), curves from individual SE models were averaged in *A*-direction:

$$A_{group}(P) = \frac{1}{n_s}\sum_{j=1}^{n_s} A_j(P) = \frac{1}{n_s}\sum_{j=1}^{n_s}\left(1 + \frac{A_{d,j}}{\gamma_j}\ln\frac{P}{P_{d,j}}\right), \quad (1)$$

where n_s is the number of subjects in each group and j is the subject number. A_j is subject j's *P-A* relationship as given by the inverse of *Eq. A3*.

For fitting purposes, we assumed that the group-averaged SE models were valid for $P \in [P_{d,group} - 15$ mmHg, $P_{s,group} + 15$ mmHg], where $P_{d,group}$ and $P_{s,group}$ are the group-averaged diastolic and systolic pressures (Fig. 1). Fitting was performed using $r_{group} = \sqrt{\frac{A_{group}}{\pi}}$ instead of A_{group} for convenience.

Fit procedure. As the used echo tracking tool (^{RF}QAS; Esaote, Maastricht, The Netherlands) considers the media-adventitia echoes of near and far walls, we assumed the measured r_{group} to reflect common carotid artery outer radius. Fitting was accomplished by varying model parameters (described below) while minimizing the sum of squared differences between model and measurements (Fig. 2). The total sum of squares ($S_{S,t}$) is the weighted sum of a radius term, one or two wall stress-related terms, and an axial force-related term, as explained below.

Circumferential wall stress was forced to be distributed homogeneously along each radial position of the wall at mean arterial pressure (MAP) (i.e., wall stress homogenization), in both the "young" and "old" situations. MAP ($P_{m,group}$) was defined as $P_{m,group} = 0.6P_{d,group} + 0.4P_{s,group}$. For the "old" group fits, average circumferential wall stress ($\bar{\sigma}_{\theta,o,MAP_0}$) at MAP could optionally be forced to remain at the "young" group level (i.e., wall stress normalization).

The normalized radius sum of squares ($S_{S,r}$) between measured (r_j) and modeled ($r_{const,j}$) radii is given by

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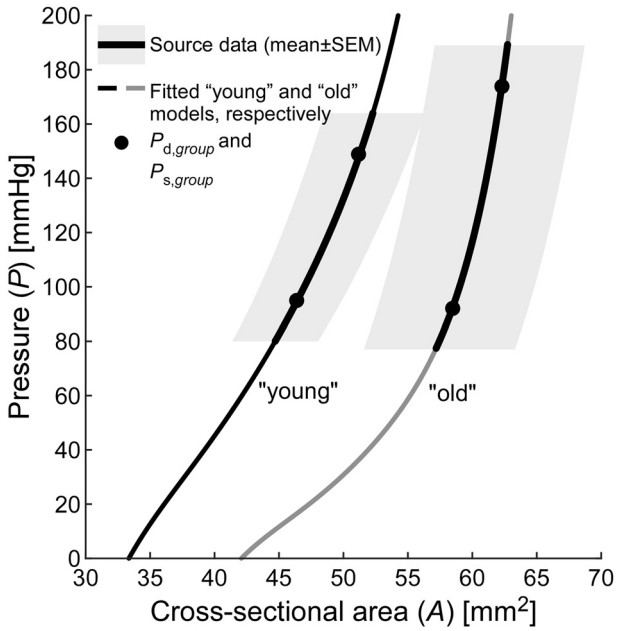


Fig. 1. Resultant constitutive model curves (thin lines) based on measured age group-specific pressure-area (*P-A*) data (thick lines). Circles indicate diastolic ($P_{d,group}$) and systolic ($P_{s,group}$) pressures in both groups. Fitting considered a pressure range of $P_{d,group}$ − 15 mmHg to $P_{s,group}$ + 15 mmHg. The "old" group curve shown was obtained by fitting the "young" constitutive model to the "old" *P-A* data under wall stress and axial force constraints (see Fig. 2*B*). Means ± SE in *A*-direction.

$$S_{S,r} = \frac{1}{n_r}\frac{1}{r_{y,d}^2}\sum_{j=1}^{n_r}\left(r_{const,j} - r_j\right)^2, \quad (2)$$

with n_r the number of data points and $r_{y,d}$ the average diastolic radius for the "young" group.

Wall stress difference ($D_{S,\sigma,n}$) is formulated as a squared difference between average wall stresses in "young" and "old" groups, both at their respective MAP:

$$D_{S,\sigma,n} = \frac{1}{\overline{\sigma}_{\theta,y,MAP_y}^2}\left(\overline{\sigma}_{\theta,o,MAP_o} - \overline{\sigma}_{\theta,y,MAP_y}\right)^2, \quad (3)$$

with $\bar{\sigma}_{\theta,o,MAP_o}$ the "old" group's average wall stress and $\bar{\sigma}_{\theta,y,MAP_y}$ the "young" group's average wall stress at MAP.

The degree of wall stress inhomogeneity is formulated as a sum of squares between local wall stress and the average wall stress:

$$S_{S,\sigma,h} = \frac{1}{n_e}\frac{1}{\overline{\sigma}_{\theta,MAP}^2}\sum_{e=1}^{n_e}\left(\sigma_{\theta,e,MAP} - \overline{\sigma}_{\theta,MAP}\right)^2, \quad (4)$$

with n_e the number of elements in which the wall is subdivided radially, $\sigma_{\theta,e,MAP}$ the local wall stress at MAP and $\bar{\sigma}_{\theta,MAP}$ the average wall stress at MAP.

Axial force was kept constant over the fitting pressure range via the degree of axial force inhomogeneity:

$$S_{S,F} = \frac{1}{n_r}\frac{1}{F_{target}^2}\sum_{j=1}^{n_r}\left(F_j - F_{target}\right)^2, \quad (5)$$

with F_j the reduced axial force at a given pressure (cf. *Eq. A23*). Reduced axial force is clearly described by Holzapfel and Ogden (24) as the force applied in the axial direction additional to that generated by the pressure on the closed ends of the tube. For the "young" group, F_{target} was 0.5 N; a value based on a canine carotid artery (37). For the "old" group, F_{target} is defined as the mean F over the wall. In other words, in the young group, reduced axial force is forced to remain constant at 0.5 N over the fitting pressure range, whereas in the old group it is forced to remain constant but at a nonfixed value.

The four sums of squares ($S_{S,r}$, $D_{S,\sigma,n}$, $S_{S,\sigma,h}$, and $S_{S,F}$) are combined into the total sum of squares ($S_{S,t}$) via

$$S_{S,t} = w_r S_{S,r} + w_{\sigma,n} D_{S,\sigma,n} + w_{\sigma,h} S_{S,\sigma,h} + w_F S_{S,F}. \quad (6)$$

w_r, $w_{\sigma,n}$, $w_{\sigma,h}$, and w_F are dimensionless weighting parameters, as specified in Table 3. Fitting was performed using the trust-region-reflective algorithm, implemented in the MATLAB Optimization Toolbox function lsqnonlin (MATLAB R2014b; The MathWorks Natick, MA). Fitting was initiated from 100 random start points in the parameter space using the MATLAB Global Optimization Toolbox function MultiStart. Radius, wall stress, and axial force fit quality were assessed using their normalized mean squared errors:

$$E_r = 100\% \cdot \frac{1}{r_{y,d}}\sqrt{\frac{1}{n_r}\sum_{j=1}^{n_r}\left(r_{const,j} - r_j\right)^2} = 100\% \cdot \sqrt{S_{S,r}}, \quad (7)$$

$$E_{\sigma,n} = 100\% \cdot \left|\frac{\overline{\sigma}_{\theta,o,MAP_o} - \overline{\sigma}_{\theta,y,MAP_y}}{\overline{\sigma}_{\theta,y,MAP_y}}\right| = 100\% \cdot \sqrt{D_{S,\sigma,n}}, \quad (8)$$

$$E_{\sigma,h} = 100\% \cdot \frac{1}{\overline{\sigma}_{\theta,MAP}}\sqrt{\frac{1}{n_e}\sum_{e=1}^{n_e}\left(\sigma_{\theta,e,MAP} - \overline{\sigma}_{\theta,MAP}\right)^2} = 100\% \cdot \sqrt{S_{S,\sigma,h}}, \text{ and} \quad (9)$$

$$E_F = 100\% \cdot \frac{1}{F_{target}}\sqrt{\frac{1}{n_r}\sum_{j=1}^{n_r}\left(F_j - F_{target}\right)^2} = 100\% \cdot \sqrt{S_{S,F}}. \quad (10)$$

Fitted model parameters. Constitutive model fitting was performed with various combinations of fixed and fitted parameters and constraints (Fig. 2 and Table 3). Fitting was first performed on the "young" *P-A* curve, yielding a take-off parameter set consisting of nine parameters for subsequent fitting of the "old" *P-A* curve (Table 3).

Residual stretch related parameters [opening angle (α) and axial prestretch (λ_z)] were fixed in the "young" situation and estimated from literature. Opening angle and axial prestretch vary markedly along the vascular tree. In this study on the carotid artery, we chose to fix "young" opening angle to 100°, based on reported carotid artery opening angles of 110° (human; Ref. 14) and 101.2° (nonhuman primates; Rhesus Macaques; Ref. 57). We fixed our "young" prestretch to 1.20, based on Delfino et al. (14) reporting $\lambda_z = 1.10$ and Wang et al. (57) reporting $\lambda_z = 1.46$.

In the "old" model, wall stress normalization was added as an additional constraint. Both the cross-sectional wall area and the zero-stress inner radius were fitted to yield the final formulation of the "old" *P-A* fitted model (Fig. 2*B*).

Arterial residual strain is known to change with age (65). Zulliger and Stergiopulos (65) derived population-based relations of aortic α and λ_z with age, based on data by Saini et al. (42). Using these formulas, we calculated aortic opening angle and axial prestretch for our "young" ($\alpha = 373°$; $\lambda_z = 1.26$) and "old" group's ages ($\alpha = 407°$; $\lambda_z = 1.19$). Notably, these values are based on measurements of the aorta and cannot be used directly when modeling the carotid artery. Therefore, subsequently, we calculated the percentage change with aging (α: +9%; $\lambda_z = -6\%$) and imposed these percentage changes onto our "old" group fits.

To assess the consistency of our measurement and modeling approach, we repeated the fitting procedure using "young" and "old" data obtained under repeat, i.e., after antihypertensive treatment, conditions.

Statistical Analyses

Statistical analyses were performed using MATLAB. Nonparametric Wilcoxon rank-sum tests were performed to evaluate statistical

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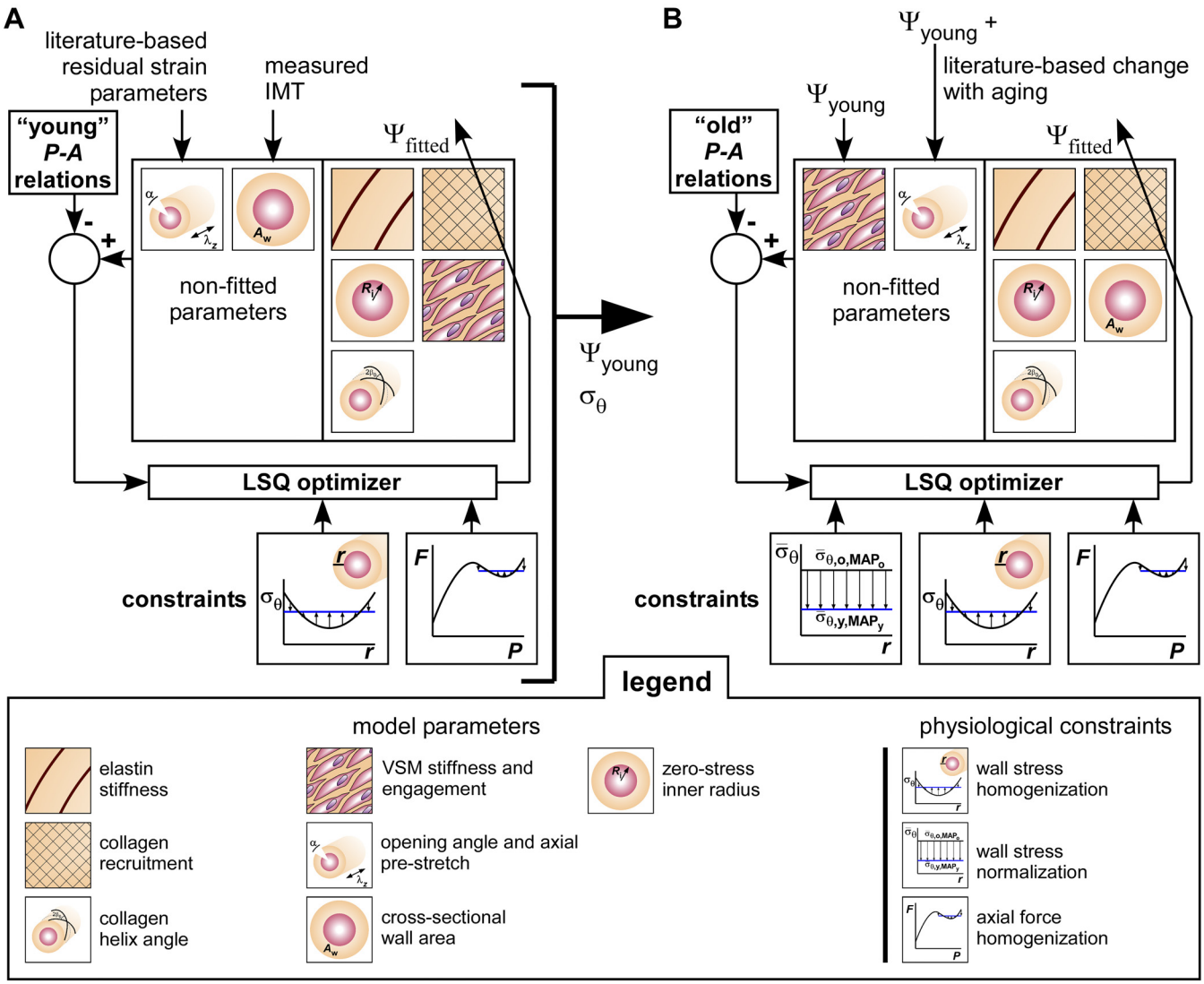


Fig. 2. Block diagrams summarizing the performed model fits on the *P-A* relations of the "young" (*A*) and "old" (*B*) patient groups. LSQ optimizer, nonlinear least-squares optimizer. A_w: cross-sectional wall area; λ_z: longitudinal stretch ratio; α: opening angle; R_i: zero stress inner radius, σ_θ: circumferential wall stress; $\bar{\sigma}_\theta$: average circumferential wall stress; $\bar{\sigma}_{\theta,o,MAP_o}$: "old" group average wall stress at mean arterial pressure (MAP); $\bar{\sigma}_{\theta,y,MAP_y}$: "young" group average wall stress at MAP; *F*: reduced axial force; IMT: intima-media thickness; VSM: vascular smooth muscle.

differences between patient groups. $P \leq 0.05$ was considered statistically significant. Unless otherwise indicated, values are given as means ± SD.

RESULTS

Source Data: Study Population and P-A Curves

Table 2 summarizes "young" and "old" group characteristics. Both groups exhibit arterial hypertension at baseline. Age groups do not show statistical differences, except in age, PWV, and compliance coefficient. Both aortic and carotid PWV in the "old" group are increased with respect to the "young" group. However, pressures also differ between groups. Table 1 shows carotid PWV for predefined pressure ranges, thereby enabling BP-independent comparison of "young" and "old" carotid stiffness. As evident from this table, PWV increases with age and with BP independently.

Figure 1 shows the *P-A* curves for the "young" and the "old" groups (thick lines). With respect to the "young" curve, the "old" curve is shifted to the right (higher cross-sectional area). Additionally, the "old" working range extends to higher pressures, effectively leading to a steeper effective *P-A* curve working area and therefore a stiffer vessel.

Constitutive Modeling

"Young" P-A fitting results. Our model was fully able to describe the "young" *P-A* relationship (Table 3, "young" column). Fit errors were small (radial fit error $E_r = 2.8 \cdot 10^{-2}\%$, wall stress homogenization error $E_{\sigma,h} = 0.34\%$, and axial force error $E_F = 0.18\%$; not shown). Note the difference in magnitude among those errors, which is caused by the difference in weighting of the radius and wall stress homogenization/axial force terms (*Eq. 6*). At MAP, elastin bears 70% of the load, whereas collagen and VSM bear the remaining 5 and 25%, respectively. Average wall stress at MAP was 81 kPa, which is in line with physiological values (9).

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Table 3. *Relative changes in constituent parameters fitting "old" group pressure-area data with "young" group fit as take-off condition*

	Fit Range			
	Lower	Upper	"Young"	"Old"
Weighting parameters				
w_r			10	10
$w_{\sigma,n}$			0	1
$w_{\sigma,h}$			1	1
w_F			1†	1
Parameter values				
Geometrical				
A_w, mm^2	10	100	14.4	*+31%*
R_i, mm	1	10	*4.0*	*+12%*
λ_z			1.20	−6%
α,°			100	+9%
Elastin				
c_{elast}, kPa	5.0	400	*181*	*−14%*
Collagen				
k	2.0	32.0	*6.7*	*0%*
b	0.2	2.1	*0.4*	*−30%*
β_0,°	0	90	*40*	*−3%*
Vascular smooth muscle				
c_{VSM}, kPa	5	100	77	*
μ	1	10.0	*3.2*	*
σ	0.05	1.0	*0.14*	*
Mean arterial pressure				
P_m, mmHg			117	125
Wall loading at MAP				
Pressure load bearing				
P_{elast}, mmHg (%)			82 (70%)	67 (54%)
P_{coll}, mmHg (%)			5 (5%)	34 (28%)
P_{VSM}, mmHg (%)			30 (25%)	23 (19%)
Collagen engagement				
CDF, %			0.58	3.5
Average wall stress				
$\bar{\sigma}$, kPa			81	81
Model wall thickness, mm			0.63	0.75
Diastolic model wall thickness, mm			0.65	0.76

Parameter percentages are 100% · ["old" fit value − "young" fit value]/["young" fit value]. Residual strain related parameters (λ_z and α) were taken from literature for the young group, as were their changes with aging (see main text). Vascular smooth muscle (VSM) parameters were evaluated in initial fits and kept constant in the final analysis, because incorporation of vascular smooth muscle parameters would have led to overfitting. MAP, mean arterial pressure; CDF, cumulative density function. Parameter values and changes in italics were obtained by means of fitting. *Parameter was kept constant at the value of the "young" group. †In the "young" group, reduced axial force (F) was, in addition to being homogenized over the fitting pressure interval, kept at 0.5 N.

Preliminary "old" P-A fitting results. To investigate the performance of our constitutive modeling approach in describing aging, we assessed several combinations of parameter changes with respect to the "young" situation (not shown). Firstly, we tried to assess aging by varying only one component's parameters (i.e., only of elastin, only of collagen, or only of VSM) and not including the wall stress homogenization constraint. Fitting was unsuccessful in all of these cases, leading to *P-A* curves that clearly deviated from the measured curves, with radial fit errors >1.5%.

We continued our study by varying the parameters of sets of two components, i.e., elastin and collagen; elastin and VSM; and collagen and VSM. Only the elastin-collagen combination yielded a successful fit. At this point, we decided to focus on using elastin-collagen changes in describing aging and not to include any additional fitted parameters. The inclusion of additional (VSM) parameters would lead to an underdetermined problem (i.e., overfitting). As such, the resulting fit would be of little value.

"Old" P-A fitting results. As indicated, we focused on elastin and collagen as the two main structural substrates of arterial aging. We added the constraint of wall stress normalization (i.e., requiring average wall stress in the "old" group to equal that in the "young" group), requiring both A_w and R_i to be included as fitting variables to yield a successful fit (Fig. 2*B*). Fitting results are given in Table 3 ("old"). Wall thickness showed a physiologically realistic increase from 0.65 to 0.76 mm (measured wall thicknesses: 0.65 and 0.79 mm). With aging, we observed a 14% reduction in elastin stiffness (c_{elast}), which, together with changes in collagen recruitment from 0.58 to 3.5%, caused load bearing to shift from elastin to collagen (elastin: from 70 to 54%, collagen: from 5 to 28%; Table 3).

Consistency of final model fitting results. The aforementioned results are based on *P-A* curves measured at baseline, with the patients on a less intensive antihypertensive regime (46). Although carotid arteries operated at clearly different operating points due to antihypertensive treatment in the repeat conditions, after 3.0 ± 0.6 mo, we did not find significant *P-A* curve changes. Repeat measurement MAPs were 100 and 107 mmHg for "young" and "old," respectively (Table A1), while corresponding baseline MAPs were 117 and 125 mmHg. We utilized these (apparently similar) *P-A* data under different hemodynamic conditions to check the consistency of our model outcomes. We obtained "young" and "old" model fits to the repeat data. Overall, results were similar to those obtained with the baseline data (Table A1). Changes in A_w, R_i, and c_{elast} with aging were +27, +11, and −11%, respectively (as compared with +31, +12, and −14% at baseline; Table 3). Changes in collagen recruitment parameters k and b, however, were markedly different at −22 and −16% (as compared with 0 and −30%). The reader is referred to the DISCUSSION for further interpretation of these differences. In both "young" and "old" fits, at baseline and repeat, collagen fiber angle (β_0) remained within the range of 35–40°, similar to Zulliger et al. (62).

Despite the difference in MAP between repeat and baseline, the age-related changes in wall loading were similar (for elastin, collagen, and VSM, respectively, −16, +23, and −6% points at baseline and −15, +22, and −6% points at repeat; Table 3 and A1). Expectedly, collagen engagement in the "old" group at repeat (2.6%) was slightly lower than at the baseline visit (3.5%; Table 3), as was average wall stress ($\bar{\sigma}_\theta$), which was 68 kPa under repeat conditions compared with 81 kPa at baseline.

DISCUSSION

Key Findings

In the present study, we used patient measurements to quantify age-related changes in carotid artery wall constituent properties and load bearing. Our model predicts that from roughly the fifth into the seventh decade, *1*) elastin stiffness decreases, *2*) load-bearing significantly shifts from elastin to collagen, and *3*) collagen recruitment increases considerably. Correspondingly, simulated diameter and wall thickness increased by about 20 and 17%, respectively. Due to the fitting

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regime, fitting the model's radius response to the measured radius response, the simulated diameter increase was equal to the measured increase. The increase in wall thickness compares reasonably well with a measured thickness increase of 21%. These geometrical changes were only consistent with patient data if wall stress regulation was considered in model fitting. Our quantitative findings remained consistent with those obtained based on repeated measurements in our patients, despite measurement noise and clear differences in BP. Our findings are well in line with the existing aging paradigm and epidemiological findings and they do suggest that age-related changes in arterial wall structure can be understood and studied quantitatively by integrating constitutive modeling and noninvasive patient data.

Previous Findings from Literature

The present study addresses key problems of previous clinically applied constitutive modeling studies. Contrary to Masson et al. (30), who used 14 parameters for fitting, our findings with respect to aging are based on changes in only 6 fitted model parameters. This significantly reduces the possible effects of overfitting. Our method uses clinically obtained, noninvasive data and can therefore easily be applied in a broad context to quantitatively assess changes in arterial wall structure and function.

Our quantitative findings on the load shift from elastin to collagen are consistent with the classic paradigm as proposed by O'Rourke and Hashimoto (36). They state that with aging the elastin substructure degrades due to wear, leading to a progressive transfer of mechanical load bearing to the collagen structure, which may be reinforced by increased crosslinking.

Comparing our findings regarding wall thickness with literature, the model-calculated wall thickness increase with aging from 41 to 64 yr (+0.11 mm) was comparable to the median increase in IMT for the same age range in the Reference Values for IMT Study (healthy subpopulation; Table 3 in Ref. 16; men: +0.131 mm, women: +0.124 mm).

With regard to vessel dilation, "young" and "old" group average vessel cross-sectional areas (50 and 60 mm^2; Fig. 1) roughly correspond to diameters of about 8.0 and 8.7 mm, respectively, supporting a near 10% increase in diameter. This compares well with the increase in proximal aortic diameter of about 13% from 23 mm at age 44 yr to 26 mm at 64 yr, as derived from data in post mortem aortic specimens at 100 mmHg (55). In our "old" fit, we forced wall stress to remain constant with age. We based this constraint on findings by Åstrand et al. (1) and Matsumoto and Hayashi (32). Åstrand et al. (1) describe carotid wall stress to remain constant with age in a human population. Matsumoto and Hayashi (32) have shown that in rats, after inducing hypertension, wall remodeling is such that wall stress at in situ BP remains constant, a finding which was confirmed by Wolinsky (61).

Influence of Wall Stress Normalization

Our "old" fit (Table 3) incorporates a wall stress normalization constraint, i.e., we force average circumferential wall stress in the "old" situation to be equal to the "young" situation. Before we enforced this constraint, we tried fitting the "old" situation without wall stress normalization, setting $w_{\sigma,n} = 0$. The results are given in Table A2. With $w_{\sigma,n} = 0$, fitting is successful if either cross-sectional area (A_w) or zero-stress inner radius (R_i) is varied. Although these fits were successful with respect to their fit errors (E_r, $E_{\sigma,h}$, and E_F for both fits), these models showed nonphysiological wall thickness changes. The fit in which A_w was fitted showed almost a doubling of wall thickness (diastolic wall thickness change from 0.65 to 1.13 mm). If R_i was fitted, wall thickness showed a decrease with age. The latter cannot be considered physiological. These results prompted us to add wall stress normalization to our "old" fits, yielding the "old" fit in Table 3.

Influence of Age-Related Changes in Residual Strain Parameters

In our "old" fit (Table 3), we have chosen to change the residual strain parameters based on the relation of aortic residual strain parameters with age as described by Zulliger and Stergiopulos (65) (see also MATERIALS AND METHODS). We assessed the influence of the age-related residual strain parameters by performing a number of additional fits. First, we fitted the "old" data while keeping λ_z and α at their young values (Table A3, "old," $\lambda_z \leftrightarrow$, $\alpha \leftrightarrow$). In this case, age-related changes in load bearing are more pronounced than in the fit with changed λ_z and α (Table 3, "old"; Table A3, "old," $\lambda_z \downarrow$, $\alpha \uparrow$). If only the increase of α is included (Table A3, "old," $\lambda_z \leftrightarrow$, $\alpha \uparrow$), load bearing changes are even more pronounced. On the contrary, if only the decrease in λ_z is incorporated (Table A3, "old," $\lambda_z \downarrow$, $\alpha \leftrightarrow$), a smaller shift in load bearing is required than in the case where both λ_z and α are varied. From the results in Table A3, we conclude that the lowering of axial stretch with age "helps" in shifting the *P-A* curve from its young to its old shape.

Influence of Age-Related Changes in VSM Parameters

In our "old" fit (Table 3), we have chosen to keep VSM-related parameters constant to prevent overfitting. However, literature reports some changes in VSM with age. Hence, we evaluated the importance of these (potential) changes on our main findings.

One influential change in VSM that may occur with age is that smooth muscle cells change their phenotype from contractile to synthetic (28). Mechanically, this may translate into a smaller number of smooth muscle cells taking part in vasoconstriction, hypothetically leading to a smaller maximum contractile force. Some studies in rat aortas indeed confirm this decrease in maximum contractile force (12, 52), whereas other studies in rat carotid artery show no decrease in maximum contractile force (15). To study the potential effect of a decrease in maximum contractile force on our aging findings, we assessed the effect of a 50% decrease in the c_{VSM} parameter with aging. Table A4 ("old," $\lambda_z \downarrow$, $\alpha \uparrow$, $c_{VSM} \downarrow$) shows that the 50% decrease in c_{VSM} removes the need for elastin destiffening. As expected, VSM load bearing in the "old" situation with reduced c_{VSM} is approximately halved compared with the situation in which c_{VSM} is kept constant.

Another change in VSM that may occur with age is related to intrinsic, passive smooth muscle cell stiffness. Qiu et al. (39) reported an increase in intrinsic smooth muscle cell stiffness of 223%, as measured by atomic force microscopy. This change may be linked to the aforementioned change to a more synthetic VSM phenotype with age. To study the effect of in-

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creased smooth muscle cell stiffness, we increased S_{basal} by a factor of 2 and assessed the influence on our aging findings. Table A4 ("old," $\lambda_z \downarrow$, $\alpha \uparrow$, $S_{basal} \uparrow$) shows that the effect of an increase of S_{basal} on our findings is negligible, also with respect to the observed reproducibility (Table A1).

Overall, we conclude that the influence of age-related changes in VSM parameters on our constitutive finding (the shift to collagen load bearing) is relatively small and that the error that we make by assuming constant VSM parameters with age is small.

Measurements, Population

Our study setup was cross-sectional. Ideally, aging would be studied longitudinally. However, we found our stiffness-BP-age pattern (Table 1) to be strikingly similar to the findings in the Reference Values for Arterial Stiffness (41). This similarity indicates that the measured cross-sectional changes in *P-A* relationships do reflect normal aging. We based our results on a relatively small population ($n = 13$). Our study, however, did evaluate the reproducibility of our approach, based on repeated measurements in the same patient but under clinically significantly different BP conditions.

We divided our study population into two, almost equally large, age groups by using an age cut-off value of 50 yr. As evident from e.g., the Reference Values for Arterial Stiffness project (41), vascular stiffening with age is a gradual, progressive phenomenon. In this light, the use of an absolute age cut-off defining "young" and "old" is inappropriate. However, given our small population, it was not feasible to use multiple age groups, which would have caused an unacceptably large measurement uncertainty.

Constitutive Modeling Assumptions

The volume fractions of collagen, elastin, and VSM in the wall were taken from literature (17, 62). Ideally, these values would be measured patient-specifically. However, using the noninvasive techniques we employed, this is not feasible. We did not fit these parameters since, in our formulation, this would lead to overfitting (also see *Fitting Approach*). For example, an increase in elastin content is mechanically indistinguishable from an increase in elastin stiffness. The same holds for the other components. Therefore, the decrease in elastin stiffness that we reported could (mechanically) also reflect a decrease in elastin content.

The mechanical response of each constituent was assumed to be hyperelastic and could therefore be described by an SEF. This approach neglects eventually present viscoelastic effects. Previously, we have shown that viscous behavior and the associated hysteresis in *P-A* relationships of the carotid artery wall are negligible in vivo when using a well-characterized measurement setup (6, 20).

Our collagen recruitment model requires some additional attention. Collagen was modeled to engage in load bearing with increased stretch. This engagement is characterized by a statistical probability density function (PDF) of the log-logistic type with a shape (k) and a scale (b) parameter. Collagen recruitment in our study, however, was (in absolute sense) very low at ~0.5–4%. This means that the only part of the PDF used is a very small part of the upslope. A shift in upslope to a lower strain (as shown in Fig. 3 when comparing the "old" and "young" curves) can, to a large extent, be accomplished both by changing k and b. In other words, k and b are not independent parameters and are therefore not uniquely identified by parameter fitting. This explains the more inconsistent behavior of k and b with the various fits.

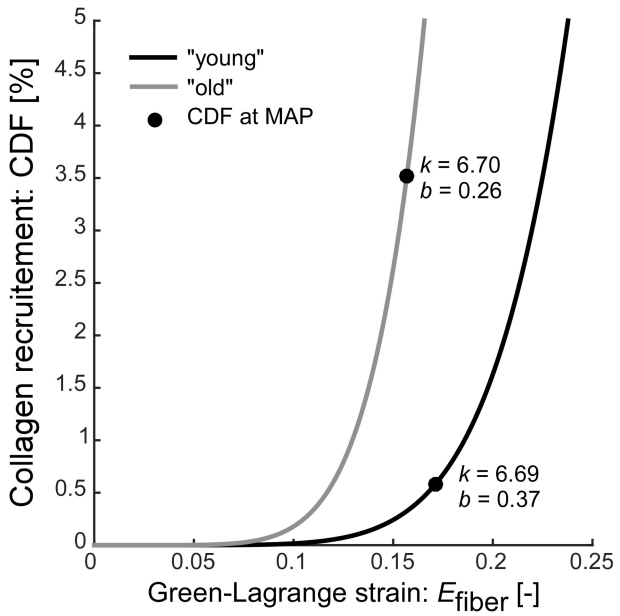


Fig. 3. Log-logistic cumulative density function (CDF) describing the cumulative collagen engagement strain distribution of the constitutive model fitted to the "young" patient group and the "old" patient group. Circles indicate the percentage of collagen engaged for the Green-Lagrange strain (E_{fiber}) at MAP. *k, b*: log-logistic shape (k) and scale (b) parameters.

In our fits, we have constrained circumferential wall stress to be homogeneous over the arterial wall. This constraint is based on reports that incorporation of residual stresses in a constitutive model greatly reduces the transmural gradient in circumferential wall stress (18, 26). We have assumed that there is a full transmural homogenization. In reality, it can be questioned whether this state is reached in an artery at working pressure. In this light, it is important to realize that we assume all wall constituents to be homogeneously distributed in the wall. Under this assumption, and given the fact that a living artery "does not know its zero-stress state" (i.e., it has, during its life span, always been subjected to the BP), it is quite likely that artery structure is such that all material contributes approximately equally to the total circumferential wall stress (18, 59). In our opinion, assuming a nonhomogeneous distribution of wall stress at working pressure, in a model with a homogeneous distribution of constituents, is therefore unrealistic.

As just mentioned, we assume that all mechanical wall constituents are homogeneously distributed in the wall. This means, e.g., that there are no separate intimal, medial, and adventitial layers and that elastin is not concentrated in layers. In a layered constitutive model (10, 25), however, wall stress distribution at working pressure would not necessarily be uniform across the wall, and our assumption of wall stress homogenization would be invalid.

Another related assumption is that the law of mixtures is obeyed in our model, which means that all wall components act mechanically in parallel. A consequence of this assumption is, e.g., that VSM contraction unloads elastin and collagen. How-

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ever, it is well possible that locally, matrix-attached smooth muscle cells load/stretch the matrix components. Modeling such effects requires highly complex models, which is beyond the scope of this article.

Fitting Approach

In nonlinear parameter fitting of (relatively) large numbers of parameters, there are two important difficulties to consider, as clearly described by Stålhand et al. (50). Firstly, there can be several local minima in the parameter space. A gradient-based algorithm (like the algorithm we used) can end in such a minimum, thereby overlooking the global minimum. We reduced this limitation of the fitting algorithm by running the algorithm from multiple start points ($n = 100$) and selecting the fitted parameter set corresponding to the lowest minimum found. Another alternative would be sampling the entire parameter space using a Monte Carlo approach, which has the disadvantage of being computationally very heavy. Secondly, a much more fundamental problem is that of overparameterization, or overfitting. In that case, the sum of squares at the optimum changes very little with varying parameters, meaning there is not a single optimal parameter combination but a range of optimal solutions for varying parameter combinations. In this case, the number of fitted model parameters needs to be reduced.

Our "young" parameter fit (varying 8 parameters) is susceptible to overfitting. However, it is not the absolute "young" parameter values that were of interest, but rather the changes in an aging hypothesis-based subset of these parameters. As described in RESULTS, when one of the constituents' parameters were fixed, fitting was unsuccessful. This is a strong indication that the young-to-old fits were not overparameterized. As we found that the elastin-collagen combination yielded a successful fit, we decided not to include additional smooth muscle related parameters, which clearly would have led to overfitting.

Perspectives

Our study demonstrates that age-related changes in arterial wall structure can be understood and studied mechanistically at the quantitative level by combining constitutive modeling and noninvasive clinical patient data. Our integrated methodology is potentially widely applicable in larger scale human studies into arterial wall remodeling. As such, our approach could be of added value in mechanistic and clinical intervention studies. However, it should be stressed that, when patient-specific assessment is considered, our approach explicitly will remain sensitive to measurement noise (as illustrated for our study in Fig. 1).

Conclusions

Our constitutive modeling approach suggests that the arterial elastic and geometric properties of older compared with younger hypertensive patients are directly linked to reduced elastin stiffness or content and advanced collagen engagement in the arterial wall, as well as to modulation of wall stress by vessel enlargement and wall thickening. We conclude that these findings could well explain the influence of aging in the stiffness-pressure-age pattern observed.

APPENDIX

P-A Curve Data Acquisition and Processing

The method of obtaining noninvasive *P-A* curves as summarized below is elaborated in (46).

Study population. The study was approved by the ethical committee of Maastricht University and conducted in accordance with the Declaration of Helsinki (updated Seoul 2008). All subjects provided written informed consent before participation. Subjects were recruited from patients referred to our outpatient hypertension clinic for a 2-day clinical assessment. In most patients antihypertensive drugs were discontinued 2 wk before the 2-day clinical assessment. Participants underwent extensive arterial function and hemodynamic measurements (detailed below) at inclusion and at 3-mo (3.0 ± 0.6 mo) follow-up (repeat). Patients were managed according to European Society of Hypertension (ESH) guidelines (29), and their treating physicians were blinded for (intermediate) study results. We identified

Table A1. *Reproducibility of findings based on repeat measurements*

	Fit Range		"Young"	"Old"
	Lower	Upper	Repeat	Repeat
Weighting parameters				
w_r			10	10
$w_{\sigma,n}$			0	1
$w_{\sigma,h}$			1	1
w_F			1†	1
Parameter values				
Geometrical				
A_w, mm^2	10	100	14.2	*+27%*
R_i, mm	1	10	*3.9*	*+11%*
λ_z			1.20	−6%
α,°			100	+9%
Elastin				
c_{elast}, kPa	5.0	400	*169.7*	*−11%*
Collagen				
k	2.0	32.0	*8.1*	*−22%*
b	0.2	2.1	*0.3*	*−16%*
β_0,°	0	90	*35*	*+11%*
Vascular smooth muscle				
c_{VSM}, kPa	5	100	*33*	*
μ	1	10.0	*3.2*	*
σ	0.05	1.0	*0.06*	*
Mean arterial pressure				
P_m, mmHg			100	107
Wall loading at MAP				
Pressure load bearing				
P_{elast}, mmHg (%)			79 (78%)	67 (63%)
P_{coll}, mmHg (%)			3 (3%)	27 (25%)
P_{VSM}, mmHg (%)			19 (19%)	13 (12%)
Collagen engagement				
CDF, %			0.34	2.6
Average wall stress				
$\bar{\sigma}$, kPa			68	68
Model wall thickness, mm			0.64	0.74
Diastolic model wall thickness, mm			0.65	0.75

This table shows the same data as Table 3 in the main text, except for the fact that "repeat" instead of "baseline" source data are used for fitting. Parameter percentages are 100% · ["old" fit value − "young" fit value]/["young" fit value]. Residual strain related parameters (λ_z and α) were taken from literature for the young group, as were their changes with aging (see main text). Vascular smooth muscle parameters were evaluated in initial fits and kept constant in the final analysis, because incorporation of vascular smooth muscle parameters would have led to overfitting. Parameter values and changes in italics were obtained by means of fitting. *Parameter was kept constant at the value of the "young" group. †In the "young" group, reduced axial force (F) was, in addition to being homogenized over the fitting pressure interval, kept at 0.5 N.

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Table A2. *Necessity of wall stress normalization with age*

	Fit Range					
	Lower	Upper	"Young"	"Old" $w_{\sigma,n} = 0$, A_w ↻, R_i ↔	"Old" $w_{\sigma,n} = 0$, A_w ↔, R_i ↻	"Old"
Weighting parameters						
w_r			10	10	10	10
$w_{\sigma,n}$			0	0	0	1
$w_{\sigma,h}$			1	1	1	1
w_F			1†	1	1	1
Parameter values						
Geometrical						
A_w, mm^2	10	100	14.4	*+85%*	*	*+31%*
R_i, mm	1	10	*4.0*	*	*+18%*	*+12%*
λ_z			1.20	−6%	−6%	−6%
α,°			100	+9%	+9%	+9%
Elastin						
c_{elast}, kPa	5.0	400	*181*	*−70%*	*+37%*	*−14%*
Collagen						
k	2.0	32.0	*6.7*	*−42%*	*+23%*	*0%*
b	0.2	2.1	*0.4*	*+13%*	*−40%*	*−30%*
β_0,°	0	90	*40*	*+4%*	*−6%*	*−3%*
Vascular smooth muscle						
c_{VSM}, kPa	5	100	77	*	*	*
μ	1	10.0	*3.2*	*	*	*
σ	0.05	1.0	*0.14*	*	*	*
Mean arterial pressure (MAP)						
P_m, mmHg			117	125	125	125
Wall loading at MAP						
Pressure load bearing						
P_{elast}, mmHg (%)			82 (70%)	38 (30%)	77 (62%)	67 (54%)
P_{coll}, mmHg (%)			5 (5%)	50 (40%)	31 (25%)	34 (28%)
P_{VSM}, mmHg (%)			30 (25%)	38 (30%)	17 (13%)	23 (19%)
Collagen engagement						
CDF, %			0.58	2.2	5.2	3.5
Average wall stress						
$\bar{\sigma}$, kPa			81	49	115	81
Model wall thickness, mm			0.63	1.11	0.56	0.75
Diastolic model wall thickness, mm			0.65	1.13	0.57	0.76

"Young" and "Old" columns are reproduced from Table 3. Parameter percentages are 100% · ["old" fit value − "young" fit value]/["young" fit value]. Residual strain related parameters (λ_z and α) were taken from literature for the young group, as were their changes with aging (see main text). Vascular smooth muscle parameters were evaluated in initial fits and kept constant in the final analysis, because incorporation of vascular smooth muscle parameters would have led to overfitting. Parameter values and changes in italics were obtained by means of fitting. ↻ indicates a fitted parameter; ↔ indicates that the respective parameter is kept at its young value. *Parameter was kept constant at the value of the "young" group. †In the "young" group, reduced axial force (F) was, in addition to being homogenized over the fitting pressure interval, kept at 0.5 N.

$n = 13$ patients whose antihypertensive treatment led to a decrease in diastolic BP of >7 mmHg, i.e., twice the a posteriori observed BP measurement variability of 3.5 mmHg. In the present analysis we used the baseline and repeat *P-A* curves of these patients. We divided the group into a "young" group (age <50 yr; $n = 6$) and an "old" group (age >50 yr; $n = 7$). The baseline characteristics of both groups are shown in Table 2.

Measurements. Arterial function measurements (total duration: 30–45 min) were performed in a quiet room (22°C) after a resting period of 15 min with subjects in supine position. Throughout the session four to eight repeated oscillometric BP readings were obtained at the left upper arm (Omron 705IT; Omron Healthcare, Hoofddorp, The Netherlands). Additionally, continuous pulsatile finger BP, heart rate (HR), and an estimated cardiac output (CO) were obtained by the Peñáz method (38) from the right middle finger (Nexfin; BMEYE, Amsterdam, The Netherlands).

Left common carotid artery diameter waveforms were obtained using a 7.5-MHz vascular ultrasound scanner (MyLab70; Esaote) operated at high frame rate as previously described (21). Diastolic diameter and distension values over six consecutive heartbeats and a real-time distension waveform display were used to judge quality of the recordings (^{RF}QAS utility; Esaote). Subsequently, left common carotid and right femoral artery tonometric pressure waveforms were obtained to assess carotid-femoral transit time (Sphygmocor; AtCor Medical, Sydney, Australia). For the calculation of aortic PWV (v_{ao}, defined as [transit distance]/[transit time]), transit distance was defined as sternal notch-to-femoral distance (distance as the crow flies, not over body surface) minus sternal notch-to-carotid distance as obtained by tape measure. Raw carotid artery tonometry waveforms were used to obtain calibrated local left common carotid artery BP waveforms (53). Further processing was performed using proprietary MATLAB code. Carotid ultrasound and arterial tonometry measurements were obtained in triplicate by a single experienced operator (J. Op't Roodt).

Waveform analysis and data processing. To enable quantitative assessment of the curvilinearity of the carotid artery *P-A* relation at individual subject level, we followed procedures similar to those described by Hermeling et al. (20). Briefly, systolic (peak), dicrotic notch, and diastolic (minimum) points were identified in the diameter (by manual cursor reading, using ^{RF}QAS) and pressure (automatic) waveforms. For diameter typically 9–12 and for pressure 18–30 heartbeats were included for each subject in each session.

Diastolic diameter was averaged over beats as recording mean and over recordings as session mean. Relative cyclic diameter variations (i.e., [systolic − diastolic]/diastolic) were averaged, rather than systolic diameters to avoid common noise. Accordingly, the relative amplitude of the dicrotic notch point (i.e., [notch − diastolic]/[systolic − diastolic]) was averaged for further analysis, rather than absolute

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Table A3. *Influence of age-related changes in residual strain parameters*

	Fit Range						
	Lower	Upper	"Young"	"Old" $\lambda_z \leftrightarrow, \alpha \leftrightarrow$	"Old" $\lambda_z \leftrightarrow, \alpha \uparrow$	"Old" $\lambda_z \downarrow, \alpha \leftrightarrow$	"Old" $\lambda_z \downarrow, \alpha \uparrow$
Weighting parameters							
w_r			10	10	10	10	10
$w_{\sigma,n}$			0	1	1	1	1
$w_{\sigma,h}$			1	1	1	1	1
w_F			1†	1	1	1	1
Parameter values							
Geometrical							
A_w, mm^2	10	100	14.4	*+31%*	*+31%*	*+31%*	*+31%*
R_i, mm	1	10	*4.0*	*+13%*	*+15%*	*+10%*	*+12%*
λ_z			1.20	*	*	−6%	−6%
α,°			100	*	+9%	*	+9%
Elastin							
c_{elast}, kPa	5.0	400	*181*	*−21%*	*−40%*	*+17%*	*−14%*
Collagen							
k	2.0	32.0	*6.7*	*−22%*	*−17%*	*+7%*	*0%*
b	0.2	2.1	*0.4*	*−28%*	*−23%*	*−38%*	*−30%*
β_0,°	0	90	*40*	*−13%*	*−10%*	*−7%*	*−3%*
Vascular smooth muscle							
c_{VSM}, kPa	5	100	*77*	*	*	*	*
μ	1	10.0	*3.2*	*	*	*	*
σ	0.05	1.0	*0.14*	*	*	*	*
Mean arterial pressure							
P_m, mmHg			117	125	125	125	125
Wall loading at MAP							
Pressure load bearing							
P_{elast}, mmHg (%)			82 (70%)	56 (45%)	49 (39%)	78 (62%)	67 (54%)
P_{coll}, mmHg (%)			5 (5%)	44 (35%)	46 (37%)	28 (23%)	34 (28%)
P_{VSM}, mmHg (%)			30 (25%)	25 (20%)	30 (24%)	19 (15%)	23 (19%)
Collagen engagement							
CDF, %			0.58	3.9	3.9	3.4	3.5
Average wall stress							
$\bar{\sigma}$, kPa			81	81	81	81	81
Model wall thickness, mm			0.63	0.75	0.75	0.75	0.75
Diastolic model wall thickness, mm			0.65	0.76	0.76	0.76	0.76

"Young" and "Old" $\lambda_z \downarrow, \alpha \uparrow$ columns are reproduced from Table 3. Parameter percentages are 100% · ["old" fit value − "young" fit value]/["young" fit value]. Residual strain related parameters (λ_z and α) were taken from literature for the young group, as were their changes with aging (see main text). Vascular smooth muscle parameters were evaluated in initial fits and kept constant in the final analysis, because incorporation of vascular smooth muscle parameters would have led to overfitting. $\lambda_z \downarrow$ indicates manual decrease of 6% in axial prestretch; $\alpha \uparrow$ indicates manual increase of 9% in opening angle; $\leftrightarrow$ indicates that the respective parameter is kept at its young value. Parameter values and changes in italics were obtained by means of fitting. *Parameter was kept constant at the value of the "young" group. †In the "young" group, reduced axial force (F) was, in addition to being homogenized over the fitting pressure interval, kept at 0.5 N.

dicrotic notch values. For carotid systolic, notch, and diastolic BP values the exact same scheme was applied. Median averaging was used throughout.

Carotid linear stiffness and compliance calculations. Carotid cross-sectional area (A) values were calculated assuming circular cross-section, using $A = \pi(\text{diameter}/2)^2$, resulting in diastolic (A_d), notch (A_n), and systolic (A_s) cross-sectional areas. Local, linear carotid PWVs (v_{car}) were calculated using the Bramwell-Hill relationship (7):

$$v_{car} = \sqrt{\frac{1}{\rho_b}\frac{P_s - P_d}{A_s - A_d}A_d}, \quad (A1)$$

with $\rho_b = 1.050$ kg/l the blood mass density and P_s and P_d the calibrated local systolic and diastolic carotid BPs. Accordingly, compliance coefficients (C) were calculated as

$$C = \frac{A_s - A_d}{P_s - P_d}. \quad (A2)$$

P-A curve description. In each individual and session, the three (diastolic, notch, and systolic) *P-A* points obtained were used to fit an established mathematical description of the *P-A* relation, i.e., an SE function (34):

$$P(A) = P_d e^{\gamma\left(\frac{A}{A_d}-1\right)}. \quad (A3)$$

γ is obtained by minimizing the sum-of-squares of differences between measured and curve notch and systolic pressures. As a line with one free parameter (γ) is fitted through two points, the line will, in general, not pass exactly through these two points.

IMT processing. Carotid ultrasound recordings were used to estimate IMT as described previously (22, 54, 60). IMTs at diastolic BP for all 19 ultrasound lines were averaged by taking the arithmetic mean for each acquisition. Subsequently, the median IMT of all acquisitions was calculated, resulting in one measured subject IMT. A group-averaged IMT was calculated by averaging (arithmetic mean) the IMTs of all subjects in one group.

Constitutive Modeling

Model assumptions. The influence of various material parameters on the *P-A* relation reconstructed by the model can give more insight in the influence of those parameters on the mechanical characteristics of the arterial wall. Throughout this study, it was assumed that the load bearing components of an artery are elastin, collagen, and VSM.

The human carotid artery was considered to be an incompressible, anisotropic thick-walled cylindrical tube (24). The elastic mechanical

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Table A4. *Influence of potential age-related changes in vascular smooth muscle parameters*

	Fit Range					
	Lower	Upper	"Young"	"Old" $\lambda_z \downarrow, \alpha \uparrow$	"Old" $\lambda_z \downarrow, \alpha \uparrow, c_{VSM} \downarrow$	"Old" $\lambda_z \downarrow, \alpha \uparrow, S_{basal} \uparrow$
Weighting parameters						
w_r			10	10	10	10
$w_{\sigma,n}$			0	1	1	1
$w_{\sigma,h}$			1	1	1	1
w_F			1†	1	1	1
Parameter values						
Geometrical						
A_w, mm^2	10	100	14.4	*+31%*	*+31%*	*+31%*
R_i, mm	1	10	*4.0*	*+12%*	*+12%*	*+12%*
λ_z			1.20	−6%	−6%	−6%
α,°			100	+9%	+9%	+9%
Elastin						
c_{elast}, kPa	5.0	400	*181*	*−14%*	*+3%*	*−17%*
Collagen						
k	2.0	32.0	*6.7*	*0%*	*+7%*	*+2%*
b	0.2	2.1	*0.4*	*−30%*	*−32%*	*−31%*
β_0,°	0	90	*40*	*−3%*	*−6%*	*−3%*
Vascular smooth muscle						
c_{VSM}, kPa	5	100	*77*	*	−50%	*
μ	1	10.0	*3.2*	*	*	*
σ	0.05	1.0	*0.14*	*	*	*
Mean arterial pressure						
P_m, mmHg			117	125	125	125
Wall loading at MAP						
Pressure load bearing						
P_{elast}, mmHg (%)			82 (70%)	67 (54%)	80 (64%)	65 (52%)
P_{coll}, mmHg (%)			5 (5%)	34 (28%)	33 (27%)	35 (28%)
P_{VSM}, mmHg (%)			30 (25%)	23 (19%)	12 (9%)	25 (20%)
Collagen engagement						
CDF, %			0.58	3.5	3.5	3.6
Average wall stress						
$\bar{\sigma}$, kPa			81	81	81	81
Model wall thickness, mm			0.63	0.75	0.75	0.75
Diastolic model wall thickness, mm			0.65	0.76	0.76	0.76

"Young" and "Old" $\lambda_z \downarrow, \alpha \uparrow$ columns are reproduced from Table 3. Parameter percentages are 100% · ["old" fit value − "young" fit value]/["young" fit value]. Residual strain related parameters (λ_z and α) were taken from literature for the young group, as were their changes with aging (see main text). Vascular smooth muscle parameters were evaluated in initial fits and kept constant in the final analysis, because incorporation of vascular smooth muscle parameters would have led to overfitting. Parameter values and changes in italics were obtained by means of fitting. *Parameter was kept constant at the value of the "young" group. †In the "young" group, reduced axial force (F) was, in addition to being homogenized over the fitting pressure interval, kept at 0.5 N. $\lambda_z \downarrow$ indicates manual decrease of 6% in axial prestretch; $\alpha \uparrow$ indicates manual increase of 9% in opening angle; $\leftrightarrow$ indicates that the respective parameter is kept at its young value; $S_{basal} \uparrow$ indicates manual increase of vascular smooth muscle basal tone parameter (S_{basal}) by a factor of 2; $c_{VSM} \downarrow$ indicates manual decrease in maximum vascular smooth muscle contraction parameter (c_{VSM}) by a factor of 2.

properties of separate elastic components of the arterial wall are expressed in terms of SEFs, based on deformation gradients (24). Additionally, a pseudo-SEF is used to describe VSM mechanics (64).

Strain energy functions. We based our analysis on SEFs, expressed in a cylindrical coordinate system (e_r, e_θ, e_z), describing the passive and active mechanical behavior of the arterial wall. Zulliger et al. (62, 64) propose SEFs that distinguish isotropic and orthotropic contributions representing respectively elastin and collagen.

ELASTIN. Elastin has been observed as a nonlinear elastic material resulting in a convex SEF for increasing elastin fiber stretch (35, 62):

$$W_{elast} = c_{elast}\left(I_1 - 3\right)^{\frac{3}{2}}, \tag{A4}$$

in which c_{elast} is the elastin elastic modulus, $I_1 = 2E_r + 2E_\theta + 2E_z + 3$ is the first invariant of the Green-Lagrange strain tensor, which, via $E_i = \frac{1}{2}(\lambda_i^2 - 1), i \in \{r, \theta, z\}$ is coupled to the three principal stretch ratios λ_i.

COLLAGEN. Collagen was assumed to be oriented in two symmetrical helices with angle $\pm\beta_0$ with respect to the circumferential direction. At low stretch values, collagen fibers appear to be in a wavy or coiled configuration in which case these fibers do not participate in load bearing (11). It is assumed that the engagement of collagen fibers when stretched is distributed in some statistical manner (64). Zulliger et al. (64) propose a log-logistic distribution (ρ_{fiber}) to describe the statistical distribution of the circumferential strain E_θ at which collagen starts bearing load. Following the method proposed by Zulliger et al. we have chosen the log-logistic probability density function to describe the engagement of collagen.

The engagement stretch PDF of collagen ρ_{fiber} is a piecewise function dependent on the Green-Lagrange strain in fiber direction E_{fiber}:

$$\rho_{fiber} = \begin{cases} 0, & E_{fiber} \le E_0 \\ \dfrac{b^k k \left(E_{fiber} - E_0\right)^{k-1}}{\left[b^k + \left(E_{fiber} - E_0\right)^k\right]^2}, & E_{fiber} > E_0, \end{cases} \tag{A5}$$

where $b > 0$ is a scaling parameter that, for increasing values, shifts the PDF to the right, causing collagen to engage at higher strains (51). At the same time, PDF height decreases and variance increases. Parameter $k > 0$ is a shape parameter, which, for increasing values, results in a higher maximum and a narrower distribution, thus causing collagen to engage more abruptly (51). E_{fiber} can be obtained from λ_{fiber} via

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$$E_{\text{fiber}} = \frac{1}{2}\left(\lambda_{\text{fiber}}^2 - 1\right), \text{ with} \tag{A6}$$

$$\lambda_{\text{fiber}}^2 = \cos^2\beta_0\lambda_\theta^2 + \sin^2\beta_0\lambda_z^2. \tag{A7}$$

Note that due to symmetry, E_{fiber} is equal for both collagen fiber families. The log-logistic PDF has a lower bound at value E_0, the Green-Lagrange strain at which collagen starts to change from a wavy configuration towards a straightened load bearing configuration. Under the assumption that collagen is unable to withstand compressive forces, we set $E_0 = 0$.

To model collagen, we distinguish between the mechanical behavior of a single collagen fiber and the mechanical behavior of the ensemble of fibers. For a single collagen fiber, an SEF is proposed (64):

$$W_{\text{fiber}} = \begin{cases} 0, & E_{\text{fiber}} \le 0 \\ \frac{1}{2}c_{\text{coll}}E_{\text{fiber}}^2, & E_{\text{fiber}} > 0, \end{cases} \tag{A8}$$

where c_{coll} is the collagen fiber elastic modulus and E_{fiber} is the Green-Lagrange strain tensor in the fiber direction.

The collagen ensemble SEF for one fiber family (at either $+\beta_0$ or $-\beta_0$) is now obtained by convolving ρ_{fiber} and W_{fiber}:

$$W_{\text{coll}}(E_{\text{fiber}}) = \int_{-\infty}^{+\infty} W_{\text{fiber}}(x)\rho_{\text{fiber}}(E_{\text{fiber}} - x)\mathrm{d}x. \tag{A9}$$

VASCULAR SMOOTH MUSCLE. In this study, we assumed that VSM is orientated in circumferential direction (19, 33). Zulliger et al. (64) propose an SEF for VSM in which for fully contracted VSM a stress-strain relationship with Cauchy stress linear in stretch ratio $\lambda_\theta^{\text{VSM}}$ results. The SEF describing the mechanical behavior of VSM is proposed as follows:

$$W_{\text{VSM}} = c_{\text{VSM}}\left[\lambda_\theta^{\text{VSM}} - \log\left(\lambda_\theta^{\text{VSM}}\right) - 1\right], \tag{A10}$$

where c_{VSM} is the VSM elastic modulus and $\lambda_\theta^{\text{VSM}}$ is the stretch ratio of VSM. Following Zulliger et al.,

$$\lambda_\theta^{\text{VSM}} = \lambda_\theta\lambda_{\text{pre}}, \tag{A11}$$

with, $\lambda_{\text{pre}} = \frac{L_{\text{r}}}{L_{\text{c}}} > 1$ accounting for the assumption that the length L_{c} of fully contracted VSM in the unpressurized state is shorter than the passive arterial components L_{r}. Following Zulliger et al., λ_{pre} is set to 1.83.

The contribution of VSM to wall mechanics is scaled by two functions, S_1 and S_2. S_1 implements strain-induced contractile behavior by varying VSM tone with deformation of the arterial wall (4, 64):

$$S_1 = S_{\text{basal}} + \left(1 - S_{\text{basal}}\right)\frac{1}{2}\left[1 + \text{Erf}\frac{Q - \mu}{\sqrt{2}\sigma}\right], \tag{A12}$$

where Erf is the error function and $S_{\text{basal}} \in [0, 1]$ is the basal muscle tone. Q is a deformation which, following Zulliger et al., is set to $Q = I_1$, implying that VSM tone is isotropically dependent on stretching of the arterial wall (44, 63). σ is the half-width of the VSM tone distribution and μ is the mean VSM engagement deformation threshold. In case the artery is not deformed ($Q = 0$), basal tone (S_{basal}) remains (5, 64). In this study, S_{basal} was set to 0.052.

S_2 assures that the modeled VSM only exerts force between certain stretch limits (64). Outside these bounds, VSM does not participate in load bearing:

$$S_2 = \begin{cases} 1, & 0.680 < \frac{\lambda_\theta^{\text{VSM}}}{\lambda_{\text{pre}}} < 1.505 \\ 0, & \text{otherwise.} \end{cases} \tag{A13}$$

OVERALL CONSTITUTIVE RELATIONSHIP. The three aforementioned SEFs are combined into a relation describing the local Cauchy stress for all three ($i \in \{r, \theta, z\}$) principal directions:

$$\sigma_i = -P_1 + f_{\text{elast}}\lambda_i^2\frac{\partial W_{\text{elast}}}{\partial E_i} + f_{\text{coll}}\lambda_i^2\frac{1}{2}\sum_{i_{\text{f}}=1}^{2}\frac{\partial W_{\text{coll},i_{\text{f}}}}{\partial E_i} + S_1S_2f_{\text{VSM}}\left(\lambda_i^{\text{VSM}}\right)^2\frac{\partial W_{\text{VSM}}}{\partial E_i^{\text{VSM}}}, \tag{A14}$$

with P_1 the local hydrostatic pressure within the wall and $W_{\text{coll},i_{\text{f}}}$ the collagen SEF of fiber family i_{f}, with $i_{\text{f}} \in \{1, 2\}$. Note that each component is multiplied by its respective volume fraction f_{elast}, f_{coll}, and f_{VSM}. This leads to the following expressions for σ_r, σ_θ, and σ_z:

$$\sigma_r = -P_1 + 3f_{\text{elast}}\lambda_r^2c_{\text{elast}}\sqrt{I_1 - 3}, \tag{A15}$$

$$\sigma_\theta = -P_1 + 3f_{\text{elast}}\lambda_\theta^2c_{\text{elast}}\sqrt{I_1 - 3} + f_{\text{coll}}\frac{\lambda_\theta^2}{\lambda_{\text{fiber}}^2}\cos^2\beta_0\tau_{\text{fiber}} + f_{\text{VSM}}c_{\text{VSM}}S_1S_2\left(\lambda_\theta^{\text{VSM}} - 1\right), \tag{A16}$$

$$\sigma_z = -P_1 + 3f_{\text{elast}}\lambda_z^2c_{\text{elast}}\sqrt{I_1 - 3} + f_{\text{coll}}\frac{\lambda_z^2}{\lambda_{\text{fiber}}^2}\sin^2\beta_0\tau_{\text{fiber}}, \tag{A17}$$

with

$$\tau_{\text{fiber}} = \lambda_{\text{fiber}}^2\frac{\mathrm{d}W_{\text{coll}}}{\mathrm{d}E_{\text{fiber}}}. \tag{A18}$$

IMPLEMENTATION. We distinguish four configurations of an artery (Fig. A1): Ω_0 is the unstressed, opened, stress-free configuration. Ω_1 is the closed configuration, Ω_2 is the closed, prestretched configuration, and Ω is the closed, prestretched, and pressurized configuration.

Taking into account the incompressibility of the tissue, relations between deformed coordinates (r, θ, z) in Ω and reference coordinates (R, Θ, Z) in Ω_0 are given by:

$$r = \sqrt{r_{\text{o}}^2 - \frac{R_{\text{o}}^2 - R^2}{\lambda_z k_\alpha}}, \theta = k_\alpha\Theta, z = \frac{l}{L}Z, \tag{A19}$$

where R_{o} and r_{o} are the outer radii in Ω_0 and Ω, respectively, k_α is a parameter relating to the opening angle (α) via $k_\alpha = \frac{2\pi}{2\pi - \alpha}$, and L and l are the lengths of the vessel in Ω_0 and Ω, respectively. Note the definitions of α and A_{w} in our study (Fig. A1), which may differ from other articles. The principal stretch ratios (λ_r, λ_θ, λ_z) with respect to the opened configuration are

$$\lambda_r = \frac{R}{rk_\alpha\lambda_z}, \lambda_\theta = \frac{k_\alpha r}{R} \text{ and } \lambda_z = \frac{l}{L}. \tag{A20}$$

For the internal (r_{i}) and external artery radius (r_{o}) in Ω, we can write:

$$r_{\text{i}} = \frac{\lambda_\theta R_{\text{i}}}{k_\alpha}, r_{\text{o}} = \sqrt{\frac{A_{\text{w}}}{\pi} + r_{\text{i}}^2}, \tag{A21}$$

where A_{w} is the cross-sectional area of the arterial wall and R_{i} is the zero-stress inner radius in Ω_0.

The pressure within the lumen of the artery is finally obtained by solving the balance equation:

$$P = \int_{r_{\text{i}}}^{r_{\text{o}}}\frac{\sigma_\theta - \sigma_r}{r}\mathrm{d}r. \tag{A22}$$

Reduced axial force is found by solving (*Eq. A23*)

$$F = \pi\int_{r_{\text{i}}}^{r_{\text{o}}}\left(2\sigma_z - \sigma_r - \sigma_\theta\right)r\mathrm{d}r. \tag{A23}$$

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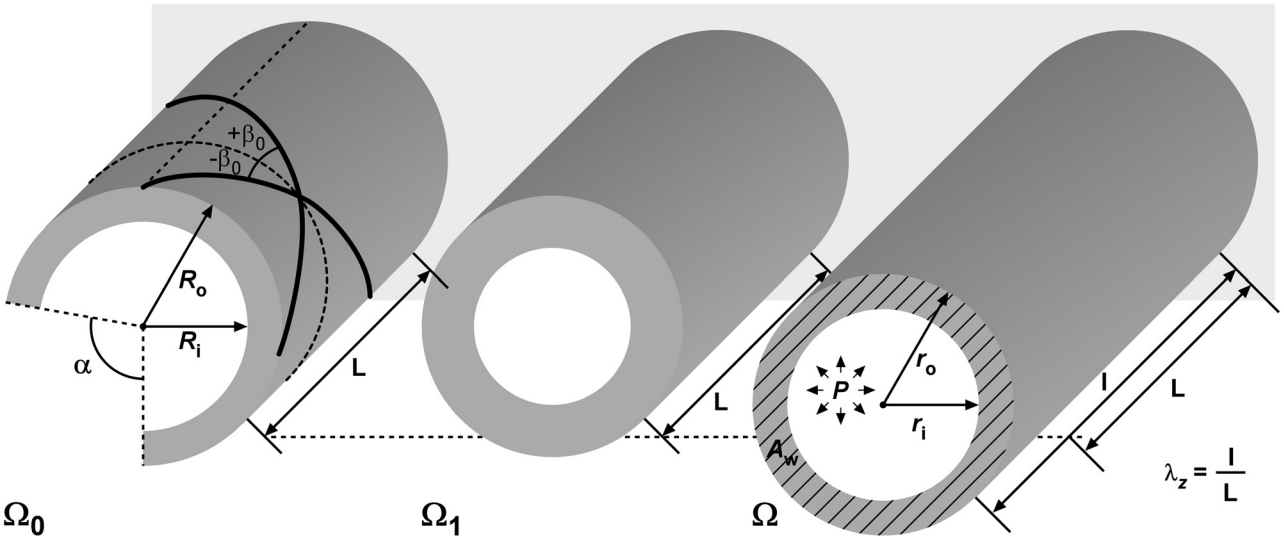


Fig. A1. Definitions of configurations. Ω_0: unstressed, opened, stress-free configuration with opening angle α, collagen fiber angle β_0, and inner and outer radii R_i and R_o, respectively. Ω_1: closed configuration. Ω: closed, prestretched and pressurized situation with cross-sectional wall A_w, luminal pressure P, and inner and outer radii r_i and r_o, respectively. Axial prestretch λ_z is defined as the quotient of unstretched (L) and stretched (l) lengths. Ω_2 (not depicted) can be obtained from Ω for $P = 0$. Note that in this study, α is defined from the center of the stress-free vessel, and that A_w is defined in Ω.

FIXED MODEL PARAMETERS. Fitted model parameters are described in MATERIALS AND METHODS. Fixed model parameters are described here. Collagen stiffness, c_{coll}, was prescribed at all simulations to be 200 MPa, which is in the range of values found in literature (27, 45).

Area fractions f_{elast}, f_{coll}, and f_{VSM} were taken as published by Fridez et al. (17, 62) for rat aorta and were, respectively, 0.306, 0.203, and 0.491.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

Author contributions: B.S., M.H.H., T.D., and K.D.R. conception and design of research; B.S., M.H.H., and J.O.t.R. performed experiments; B.S., M.H.H., A.G.d.L., T.D., and K.D.R. analyzed data; B.S., M.H.H., W.P.D., T.D., and K.D.R. interpreted results of experiments; B.S. and M.H.H. prepared figures; B.S., M.H.H., and K.D.R. drafted manuscript; B.S., M.H.H., W.P.D., A.A.K., T.D., and K.D.R. edited and revised manuscript; B.S., M.H.H., W.P.D., A.G.d.L., J.O.t.R., A.A.K., T.D., and K.D.R. approved final version of manuscript.

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